

commonly used user IDs and passwords, with numeric values and special characters; and finally, large databases of credentials already used by various people. While trying every possible combination until one is accepted is exhaustive work, with improved computing power and network bandwidths, this has become easier to do. Currently, attackers also use credential hashes, called 'rainbow tables', to further reduce break-in time.

- **Session eavesdropping:** Since authentication involves a challenge or response between two systems, credentials pass over the wire. Attackers can use packet-capturing tools while authentication is in progress, to intercept sessions and decipher credential information. For example, websites that use simple HTML forms of authentication without SSL may have clear-text passwords that are simple to capture. In a more advanced form of the same intrusion, attackers can use a session-takeover attack, where the valid token received after authentication is captured and used to impersonate the victim.
- **Replay attack:** A bit different from a session takeover; here the attacker simply records the data of a successful authentication, and initiates new requests to the server or verifier, replaying the recorded information for authentication and to impersonate the victim.

Network Authentication	Web Authentication
• NTLM • Negotiate • Kerberos • CHAP	• Basic • Digest • Form Based • Client Certs

Figure 2: The various types of authentications

- **Cookie intercept:** Web portals often rely on client-side (browser) cookies. While these should only have harmless information, such as Web browsing history, screen theme colours, etc, often the user's authentication token (issued after log-in) is stored in cookies too. Since cookies are presented by browsers to the Web server in each Web request, they can be intercepted on the wire, or using a Trojan on the browser side. Once the token is captured, the attacker can use it to impersonate the victim.
- **Verifier impersonation:** We know that MITM attacks are stealthy and very difficult to detect. Advanced

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